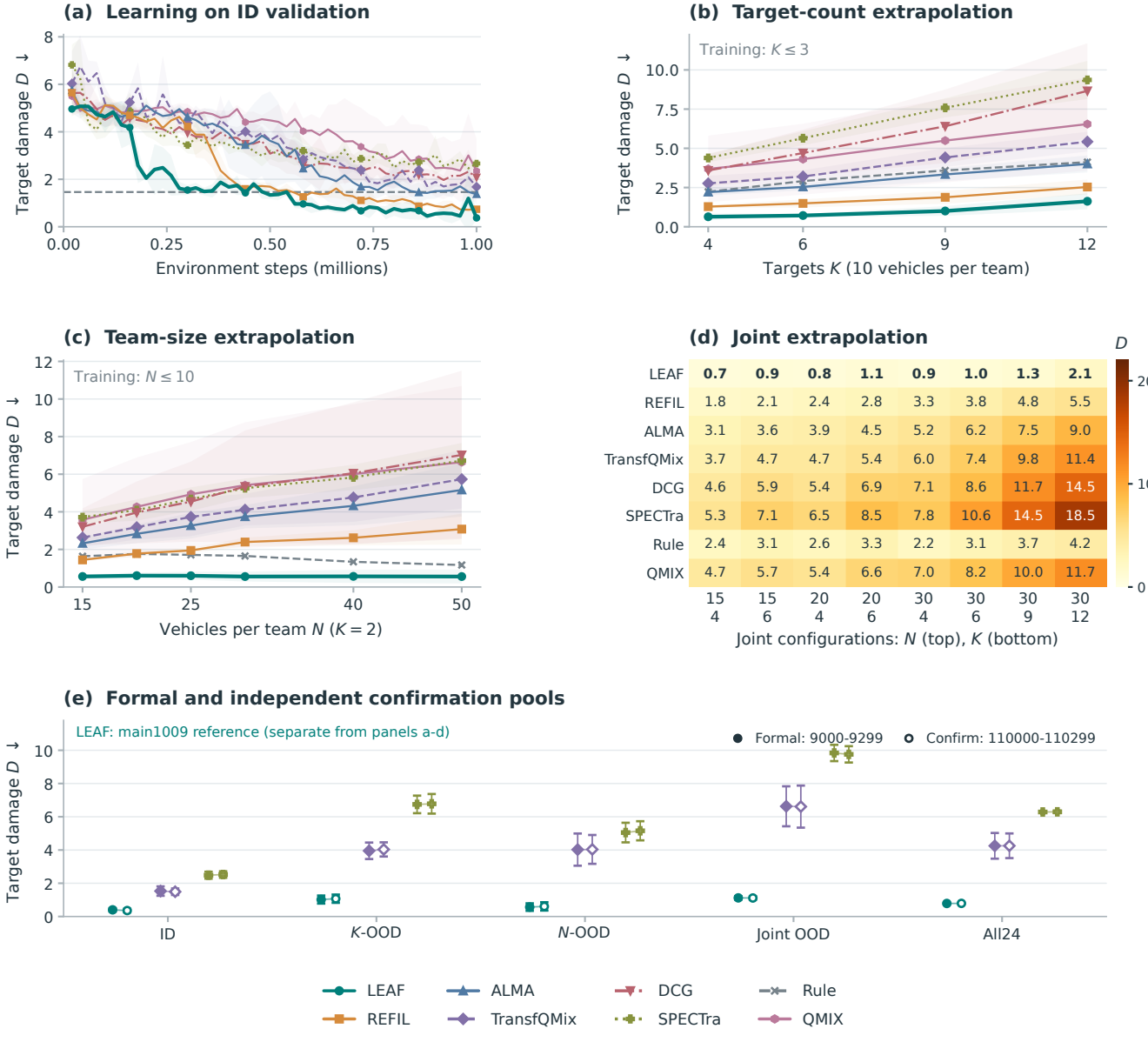

(a) Learning on ID validation
Target damage D ↓
Environment steps (millions)
(b) Target-count extrapolation
Training: K ≤ 3
Target damage D ↓
Targets K (10 vehicles per team)
(c) Team-size extrapolation
Training: N ≤ 10
Target damage D ↓
Vehicles per team N (K = 2)
(d) Joint extrapolation
D
LEAF 0.7 0.9 0.8 1.1 0.9 1.0 1.3 2.1
REFIL 1.8 2.1 2.4 2.8 3.3 3.8 4.8 5.5
ALMA 3.1 3.6 3.9 4.5 5.2 6.2 7.5 9.0
TransfQMix 3.7 4.7 4.7 5.4 6.0 7.4 9.8 11.4
DCG 4.6 5.9 5.4 6.9 7.1 8.6 11.7 14.5
SPECTra 5.3 7.1 6.5 8.5 7.8 10.6 14.5 18.5
Rule 2.4 3.1 2.6 3.3 2.2 3.1 3.7 4.2
QMIX 4.7 5.7 5.4 6.6 7.0 8.2 10.0 11.7
15 15 20 20 30 30 30 30
4 6 4 6 4 6 9 12
Joint configurations: N (top), K (bottom)
(e) Formal and independent confirmation pools
LEAF: main1009 reference (separate from panels a-d)
Formal: 9000-9299
Confirm: 110000-110299
Target damage D ↓
ID
K-OOD
N-OOD
Joint OOD
All24
LEAF
REFIL
ALMA
TransfQMix
DCG
SPECTra
Rule
QMIX